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A geographically weighted regression kriging approach for mapping soil organic carbon stock

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ABSTRACT

Local variations in the model parameters can play an important explanatory role in the spatial modeling of soil organic carbon (SOC) stock. Linear regression models assume parameters to be spatially invariant and are unable to account for the spatially varying relationships in the variables. A recently developed approach, geographically weighted regression kriging (GWRK), was used in this study to examine the relationships between environmental variables and SOC stock for the state of Pennsylvania, USA. The specific objectives were to (i) estimate the SOC stock (kg C m^{-2}) to 1.0-m depth, and (ii) compare the GWRK results with those obtained from regression kriging (RK). Data for 878 georeferenced soil profiles, extracted from National Soil Survey Center database, were divided into calibration ($n=702$) and validation ($n=176$) datasets. Environmental variables including temperature, precipitation, elevation, slope, geology, land use, and normalized difference vegetation index were explored and included as independent variables to establish the model for estimating the SOC stock. Results using Pennsylvania as a case study conclude that GWRK was the least biased and more accurate compared to RK for estimating the SOC stock based on the lowest root mean square error (2.61 vs. 4.61 kg m^{-2}), and high R^2 (0.36 vs. 0.23) values. Higher stock was consistent with higher precipitation and cooler temperature of the region. Total SOC stock ranged from 1.12 to 1.18 Pg for the soils of Pennsylvania. Forests store the highest SOC stock (64% of the total), followed by croplands (22%), wetlands (2.3%), and shrubs (2%). Results show that GWRK enhances the precision for estimating the SOC stock compared to the RK since the former takes into account the spatial non-stationarity coupled with spatial autocorrelation of the residuals.

1. Introduction

Soil is the largest terrestrial carbon (C) pool in the biosphere, storing 1.5 to 3 times more C compared to that stored in vegetation (Wang et al., 2004) or in the atmosphere. Globally, soil C stock includes about 1500 Pg (1 Pg $=10^{15}$ g) of soil organic carbon (SOC) and 950 Pg of soil inorganic carbon (Lal, 2004). Small changes in the SOC stock may influence the atmospheric CO_2 concentrations and in turn impact the global climate (Lal, 2003). Therefore, estimations of this stock are very important to explore the areas with higher or lower potential of SOC stock so that effective land use management decisions can be implemented.

Predicting soil attributes at unsampled locations using different geographic or purely spatial approaches has been used since late 1960s (McBratney et al., 2003). These predictions, however, have some uncertainties due to natural soil variability and lack of extensive soil sampling data (Batjes, 1996; Post et al., 1990; Wang et al., 2004). With the advent of geographical information system (GIS) and remote sensing, high resolution maps of climate, topography, texture, soil type and land use are easily derived and used for the estimations. As SOC stock is strongly affected by environmental variables (Minasny et al., 2006), these variables have been widely explored to estimate the SOC stock across a region (Hunt et al., 2003).

Estimation approaches that incorporate environmental variables have better prediction performance compared to generic geostatistical models such as ordinary kriging (OK) (Bishop and McBratney, 2001). Numerous studies have shown the correlative relationships between environmental variables and SOC for mapping the spatial distributions of the stock (e.g., Odeh et al., 1995; Rawlins et al., 2009). The most commonly used approaches in these studies include multiple linear regression (MLR), OK, and regression kriging (RK). There are a few limitations in these approaches that need to be addressed. Residuals in the global regression models are assumed to be identically and independently distributed but in some cases actually show strong spatial autocorrelation. Regression Kriging combines the trend fitted by global regression with the kriged residuals, which makes better use of the available data and thereby improving the accuracy of the estimations compared to MLR (Karl, 2010) and OK. Several studies have indicated that the geostatistical hybridized approaches, such as RK, perform better than the simple MLR or OK approaches (e.g. Goovaerts, 1997; Hengl et al.,

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2007; Odeh et al., 1995). However, all of these approaches have not been adapted to fit data locally with arbitrary neighborhoods for the regression (Walter et al., 2001). Other models such as regression trees have also been used for predicting the continuous soil attributes such as sand and silt contents, and water retention from terrain attributes (slope, curvature) by Pachepsky et al. (2001). These models have the ability to deal with nonlinearity. However, one of the limitations of regression trees models are the discrete predictions from each terminal node (McBratney et al., 2003). Hence, to overcome some of the above mentioned limitations, some form of non-stationary (a relationship that varies with the absolute location across the space) model is needed (Lloyd, 2010).

Since the late 1990's, a simple approach called geographically weighted regression (GWR) was developed by Brunsdon et al. (1996) and Fotheringham et al. (2002). Some of the applications of this approach have been to (i) explore the spatial variations in the relationships among variables, (ii) identify, map and model the complex spatial variations in parameter estimates (Brunsdon et al., 1996; Fotheringham et al., 2002), (iii) model spatially heterogeneous processes (e.g. Brunsdon et al., 1996; Fotheringham et al., 2002; Lloyd, 2010; Páez et al., 2008), (iv) estimate the regional SOC stock (Mishra et al., 2010), and (v) investigate the spatiotemporal change of forest net primary production with the maximum normalized difference vegetation index (Wang et al., 2008). GWR is an extension of the traditional regression framework and in contrast to global models allows varying coefficients for the environmental covariates at different locations (Brunsdon et al., 1996). Localized coefficients of the GWR are based on a weighting matrix in which observations around a sample point are weighted using a distance decay function, meaning that more proximal observations have a greater influence on the resulting localized regression coefficients (Kupfer and Farris, 2007). This approach is based on spatial non-stationarity which provides flexible estimations of the parameters (e.g. Brunsdon et al., 1996; Fotheringham et al., 2002). Any relationship that is not stationary over space unable to represent by a global statistic (Fotheringham et al., 2002). This global value may be very misleading locally, as most of the spatial variability in the SOC stock results from the local (small-scale) environmental variables, and hence cannot be explained by the global regression trend (Lado et al., 2008). GWR uses the spatially varying regression parameters to estimate values at unsampled locations but does not directly consider spatial dependence in the process of model development.

Geographically weighted regression kriging (GWRK) is a hybrid technique and an extension of the GWR approach. GWRK is composed of two components, deterministic and stochastic, where both are modelled separately. The deterministic component is modelled with GWR and uses available covariate information to predict the trend of a target variable. The stochastic component (residuals) can be interpolated with kriging and added to the estimated trend. Residuals are considered the errors, and it is possible that the errors have some spatial correlation structure that can be modelled. It could be considered that the errors are the component of the model which could not be explained by the deterministic component but is important to add as it helps explaining the variation of the target variable across space. GWRK approach can be used for estimating the residuals for predicting the trend more efficiently (Harris et al., 2010). In this context, such an approach is applied with the specific objectives: (i) to estimate the SOC stock up to 1.0 m depth for the soils of Pennsylvania, (ii) to compare the GWRK estimates with those obtained from RK, and (iii) to calculate the SOC stock stored in different land uses of the study area.

2. Materials and methods

2.1. Study area and environmental variables

The study was conducted for the state of Pennsylvania (PA) which lies between 39° 43′ to 42° 16′ N latitude and 74° 41′ to 80° 31′ W longitude. The geographical area of PA is 117,599 km^2 and elevation ranges between 6 and 1050 m a.m.s.l. The state is comprised of 11 major land resource areas (MLRAs) (Fig. 1). The MLRA is defined as the geographical units with similar patterns of climate, water resources, soils, and land uses (SCS, 1981). The soils of PA are the product of the general forces of weathering, vegetation, climate, and time working on the particular parent material and topography of the region (Pensylvania State University, 1982). The most frequent soil orders present in the study area are Alfisols, Ultisols, and Inceptisols and few Mollisols (NRCS, 2006).

A total of 878 georeferenced data points were extracted from the National Soil Survey Characterization Lab Database (NSSC, Unpublished), and divided randomly into calibration (80%; $n = 702$) and validation (20%; $n = 176$) datasets. Locations of the calibration and the validation sites in different MLRAs of the state are shown in Fig. 1.

Environmental variables used in this study for the estimation of SOC stock (kg m^{-2}) were: slope gradient, elevation, hillshade, mean annual air temperature (MAAT), mean annual precipitation (MAP), geology and land use maps, and normalized difference vegetation index (NDVI). The NDVI is described as:

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}} \quad (1)$$

where NIR is the near-infrared band and red is the red band (Rouse et al., 1973). NIR and Red in Eq. (1) correspond to band 4 and 3, respectively, in Landsat ETM. The NDVI values for the present study were scaled as: $(\mathrm{NDVI} \times 200) + 50$, which yields values from 0 to 250 (Mishra et al., 2010). The terrain attributes of elevation and slope gradient were calculated using WhiteBoxGAT (version 1.0.7, University of Guelph, Ontario, CA).

Digital elevation model (DEM), land use and geology maps of the study area were extracted from the U.S. Geologic Survey database. The NDVI data were extracted from the Global Land Cover Facility database. Climatic data (MAAT and MAP) were extracted from the database of Spatial Climatic Analysis Service of the Oregon State University (Daly et al., 2001). Note that the climatic data are the long-term average of the 1970 to 2000 period. The U.S. Geologic Survey database shows that a total of more than 40 different types of bedrock and ~15 types of land use types are present in the state of Pennsylvania. The land use types were further reclassified into six categories: (1) forest (deciduous, evergreen and mixed forests), (2) shrubs (shrubs and grasslands/herbaceous), (3) cropland (pasture/hay and cultivated crops), (4) wetland (woody wetland, emergent herbaceous wetlands and open water), (5) developed (open space, low, medium and high intensity developed areas), and (6) barren. The spatial resolution of all the environmental variables was 30 m. Resampling was used to conform any of the variables that were not resolved to the common 30 m grid.

2.2. Calculation of SOC stock

The SOC stock for each profile was calculated by summing up the SOC concentrations (kg C kg^{-1}) in each soil horizon from the surface to 1.0-m depth as follows (Eq. 2):

$$C_{stock} = \sum_{i=1}^{n} (C_i \times \rho_b \times d_i) \quad (2)$$

where C_{stock} is the SOC stock (kg m^{-2}) up to 1.0-m depth, i is the soil horizon (1, 2, 3…n), C_i is the SOC concentration (kg kg^{-1}) in the ith horizon, ρ_b is the soil bulk density (kg m^{-3}), and d_i is the thickness of the ith horizon (m). A pedotransfer function (PTF) was used to estimate the ρ_b for those pedons where it is not provided by the NSSC database. The PTF, developed by Calhoun et al. (2001), was used for calculating the soil bulk density. Soil ρ_b was estimated differently by the Adams (1973) method for the pedons having SOC concentrations

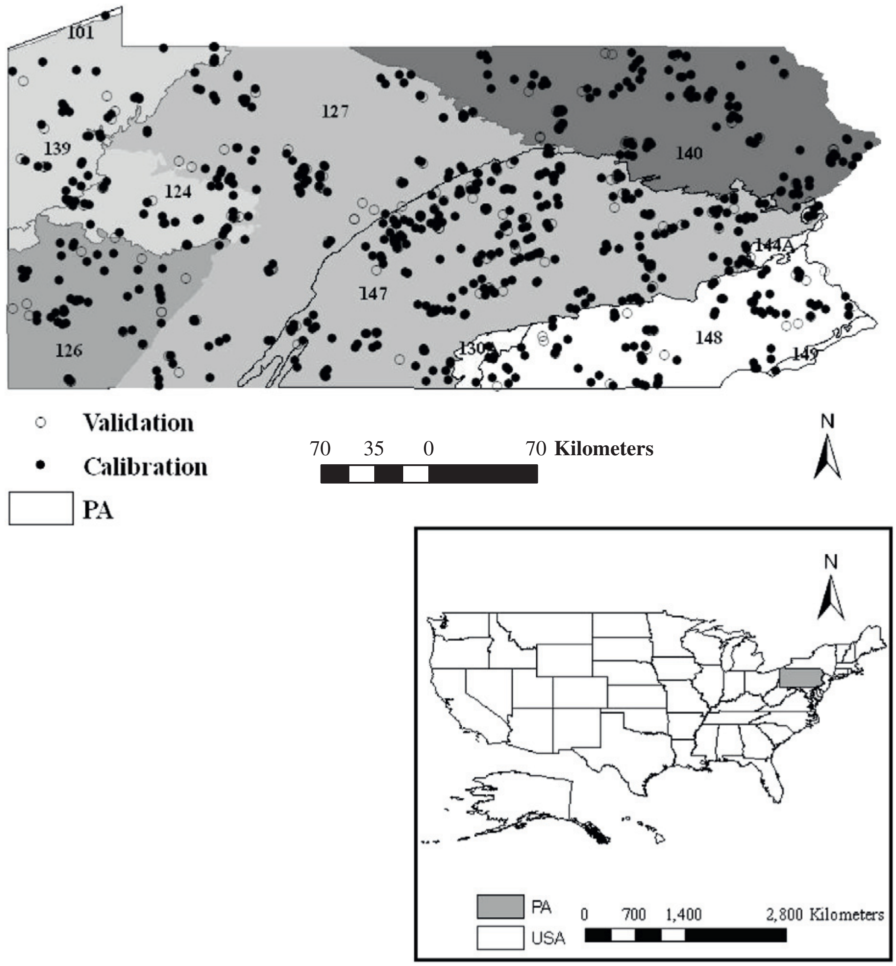

Fig. 1. Calibration and validation locations in major land resource areas (MLRA) of Pennsylvania (PA). Inset: The location of PA in U.S.A. Note: Numbers shows in the map are MLRAs present in the state.

of $>6\%$. The total SOC stock was calculated by multiplying SOC ($\mathrm{kg\ m^{-2}}$) with the land area ($\mathrm{m^2}$).

2.3. Geographically weighted regression kriging (GWRK)

The GWRK is an extension of GWR, hence the latter is discussed first in this section. Geographically weighted regression (Brunsdon et al., 1996; Fotheringham et al., 2002) extends the traditional regression by allowing local rather than global parameters to be estimated, and takes the spatial locations of data points into consideration (Zhang et al., 2011). This is an advanced and recent approach for modeling spatially heterogeneous processes. The GWR model used for the present study is shown below in Eq. (3):

$$\hat{C}_{gwr}(s_0) = \textstyle\sum_{k=0}^{p} \hat{\beta}_k(s_0) \times X_k(s_0) \qquad (3)$$

where $\hat{C}_{gwr}(s_0)$ is the predicted SOC stock at location s_0, $X_k(s_0)$ are the kth independent variables with the value X_k at location s_0, $\hat{\beta}_k$ are the unknown regression coefficients that are spatially variant, p is the total number of environmental variables. Whereas, the MLR model can be described in Eq. (4) as below:

$$\hat{C}_{mlr}(s_0) = \sum_{k=0}^{p} \hat{\beta}_k \cdot X_k(s_0) \qquad (4)$$

where $\hat{C}_{mlr}(s_0)$ is the estimated SOC at location s_0, $X_k(s_0)$ are the environmental variables, $\hat{\beta}_k$ are the estimated regression coefficients, p is the number of predictors or environmental variables. This type of model is aspatial, meaning no geographical location information is considered in the estimation of model parameters, and all of these parameters are the average across the whole dataset (Li et al., 2010). In MLR, the coefficients are assumed be spatially invariant.

Similar to Zhang et al. (2011), an adaptive spatial kernel was used in the model analysis for the present study so that the spatial extent for samples included in the GWR varied based on sample density. GWR analysis was performed using GWR 3.0 software. The latter allows selection of an optimal bandwidth (i.e. a fixed number of nearest neighbors, with a smaller bandwidth in areas that are densely sampled and a larger bandwidth in those areas that are sparsely sampled). All the calibration data were included for the bandwidth selection. Additional information related to GWR is well documented by various researchers (e.g. Fotheringham et al., 2002; Harris et al., 2010; Lloyd, 2010). The stepwise regression analysis was used to identify significant environmental variables that affect SOC stock. This analysis was performed using the SAS software (version 9.2; SAS, 2007). Stepwise regression ensures that only significant covariates are included in the final spatial regression model. A stepwise regression also expedites the process and facilitates its practical application by avoiding testing a potentially large number of covariate combinations (Yang and Jin, 2010).

For the GWRK approach, residuals from the GWR were kriged to the prediction grid using the isotropic variogram model parameters, and the kriged residual map was added to the regression predicted map for obtaining GWRK map. The equation used to perform GWRK is shown below in Eq. (5):

$$\hat{C}_{gwrk}(s_0) = \hat{C}_{gwr}(s_0) + \hat{\varepsilon}_{ok}(s_0) \tag{5}$$

where $\hat{C}_{gwrk}(s_0)$ is the SOC stock at location $s_0, \hat{C}_{gwr}(s_0)$ is the drift fitted using GWR as shown in Eq.(3), and $\hat{\varepsilon}_{ok}$ are the residual values interpolated with OK. The exponential variogram was used for the present dataset for the OK prediction. The exponential variogram model is described in Eq. (6) as below:

$$\gamma(h) = c_0 + c[1-e^{(-h/r)}] \tag{6}$$

where $\gamma(h)$ is the semivariance at lag distance h, c_0 is the nugget effect, c is the partial sill, and r is the effective range, the distance at which γ equals 95% of the sill variance (c_0+c), approximately $3r$. The slope of effective range at the origin is c/r. The *Surfer 9* software (Golden Software, Inc., Golden, Colorado) was used for developing the variogram graphs.

If the residuals from MLR exhibit spatial dependence, an estimate of the value of the regression residuals at the unknown locations can be made using OK. This process was referred to as the RK by Odeh and McBratney (1994); Odeh et al., 1994). For RK approach, trend component (deterministic, explained by the regression model) and residuals (stochastic part) can be fitted separately, and the results summed together to obtain the spatial predictions. The RK is based on a standard multiple regression and the equation used to perform this approach is given below (Eq. 6):

$$\hat{C}_{rk}(s_0) = \hat{C}_{mlr}(s_0) + \hat{\varepsilon}_{ok}(s_0) \tag{6}$$

where $\hat{C}_{rk}(s_0)$ is the estimated SOC stock at location $s_0, \hat{C}_{mlr}$ is the fitted drift using linear regression analysis of Eq. (4), and $\hat{\varepsilon}_{ok}$ are the residual values interpolated with OK. The residuals and regression predicted maps were added to obtain the final RK map. Similar environmental variables were used for retaining the comparability of the two geostatistical (GWRK and RK) approaches.

2.4. Methods evaluation

The performance of GWRK was compared with the RK approach. A total of 176 reserved (validation datasets) SOC samples were used for evaluating the performances of the studied spatial interpolation approaches. The estimated SOC stock was compared with the measured values based on mean estimation error (MEE), mean absolute estimation error (MAEE), and root mean square error (RMSE) as described in Eqs. (7)–(9), respectively.

$$\mathrm{MEE} = \frac{\sum_{i=1}^{n}\left[\hat{C}(s_i) - C(s_i)\right]}{n} \tag{7}$$

$$\mathrm{MAEE} = \frac{\sum_{i=1}^{n}\left|\hat{C}(s_i) - C(s_i)\right|}{n} \tag{8}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left[\hat{C}(s_i) - C(s_i)\right]^2}{n}} \tag{9}$$

where $\hat{C}(s_i)$ is the estimated SOC stock at s_i location, $C(s_i)$ is the observed SOC stock at location s_i, and n is the total number of sample observations. The approach with the lowest RMSE, MAEE and MEE values was used for calculating the SOC stock for the study area.

Table 1
Descriptive statistics of measured soil organic carbon stock, log-transformed C (LnC), elevation (Elev), precipitation (Ppt), and temperature (Temp) of the study area (n = 848).

Variables	SOC[a] kg m^{-2}	LnC kg m^{-2}	Elev m	Ppt mm	Slope °	Temp °C
Mean	10.6	2.11	355	1114	5.71	9.23
Median	7.57	2.02	351	1095	4.17	9.29
Minimum	1.11	0.10	6	885	0.014	6.54
Maximum	167.9	5.12	1050	1395	30.7	12.9
Standard deviation (σ)	12.2	0.62	164.5	93.7	5.17	1.37
CV (%)	115.0	29	46	8	91	15
Coefficient of kurtosis	56.9	5.60	2.86	2.89	6.41	2.11
Coefficient of skewness	6.27	0.92	0.44	0.43	1.74	0.02

[a] SOC is the soil organic carbon stock; LnC, log-transformed soil organic carbon stock.

3. Results and discussion

3.1. Descriptive statistics

The SOC stock varied from 1.11 to 168 kg m^{-2} with an arithmetic mean and median of 10.6 and 7.57 kg m^{-2}, respectively (Table 1). Statistical distribution of the measured SOC stock was positively skewed and had a sharp peak (positive kurtosis; Table 1). The coefficient of kurtosis was high with a value of 56.9 kg m^{-2}, indicating that the distribution of SOC stock data have more peaked values compared to those of the normal distribution. The SOC stock data were transformed to lognormal to create an approximately normal distribution, with mean (2.11 kg m^{-2}) and median (2.02 kg m^{-2}) being about the same. Skewness of the lognormal SOC stock was reduced from its original value of 6.27 to 0.92 kg m^{-2}, and kurtosis from 56.9 to 5.60 kg m^{-2}. The geometric mean of SOC was 2.11 kg m^{-2}. Since the SOC has the lognormal feature, geometric mean and median are more representative of the mean value of SOC compared to an arithmetic mean (Liu et al., 2006). Therefore, for estimating the SOC, lognormal transformed SOC (LnC) stock was used because the lognormal transformations can be useful in providing a more representative estimated value by reducing the variation between extreme values.

Table 1 also shows the basic statistical values of the environmental parameters of the study area. The mean values of elevation,

Table 2
Summary of the coefficients used in geographically weighted regression model, and Moran's index and p-values of the coefficients.

Variables	Minimum	Maximum	Median	Moran's index	p-value
Intercept	−9.92	19.96	1.83	0.68	0.01
Slope	−0.06	0.07	0.00	0.79	0.01
Temperature	−1.05	0.52	−0.04	0.67	0.01
Precipitation	−0.01	0.01	0.00	0.63	0.01
Elevation	−0.01	0.01	0.00	0.67	0.01
NDVI[a]	−0.01	0.01	0.00	0.90	0.01
Hillshade	−0.73	0.74	0.11	0.68	0.01
Wetland	−1.47	3.22	0.00	0.48	0.01
Cropland	−1.27	0.61	−0.05	0.72	0.01
Forest	−0.65	0.81	0.07	0.85	0.01
Limestone	−1.11	1.35	0.09	0.51	0.01
Argillaceous limestone	−1.11	0.00	0.00	0.73	0.01
Gravelly sand	−1.81	0.05	0.00	0.62	0.01

[a] NDVI, normalized difference vegetation index.

Table 3
Environmental variables used in the global regression model.

Predictor variable	Model coefficient	Error	t value	p-value
Intercept	2.8239	0.4324	6.53	0.00
Slope	−0.0014	0.0041	−0.35	0.05
Temperature	−0.0962	0.0264	−3.65	0.00
Precipitation	0.0001	0.0002	0.67	0.05
Elevation	0.0005	0.0002	2.14	0.03
NDVI[a]	−0.0013	0.0011	−1.22	0.02
Hillshade	0.1262	0.0581	2.17	0.03
Wetland	0.4598	0.2132	2.16	0.03
Cropland	−0.1357	0.0670	−2.03	0.04
Forest	0.0501	0.0666	0.75	0.04
Limestone	0.2223	0.0712	3.12	0.00
Argillaceous limestone	−0.5911	0.4124	−1.43	0.04
Gravelly sand	−0.8538	0.4171	−2.05	0.04

[a] NDVI, normalized difference vegetation index.

precipitation, slope and temperature are 355 m, 1114 mm, 5.71°, and 9.23 °C, respectively.

3.2. Estimation of the SOC stock

Summary statistics of coefficients used in the GWR approach, and parameters and their probability values used in MLR for estimating the SOC stock are shown in Tables 2 and 3, respectively. A total of 12 environmental variables were significantly ($P<0.05$) correlated with the SOC stock, and are thus used in the estimation process. The information of Moran's Index (I) and probability values of different coefficients are also included in Table 2. The Moran's I values range between 0.48 and 0.90, and show the significant spatial autocorrelation in the coefficients (Table 2). The GWR coefficients could be negative or positive, showing that the relationship between SOC stock and environmental covariates was different at different locations.

The experimental variogram of LnC, residuals of MLR (residuals$_{MLR}$), and residuals of GWR (residuals$_{GWR}$) are shown in Fig. 2. The best-fit variogram model used for the present study was exponential. Spatial dependence structure was strong with low nugget/sill ratios for LnC (38%), residuals$_{MLR}$ (21%), and residuals$_{GWR}$ (37%; Table 4). The nugget/sill ratio of LnC reveals that SOC stocks to 1.0 m depth are spatially dependent and that 38% of SOC stocks variability consists of unexplainable or random variations (Table 4). Lower ratios represent the strong spatial dependence (Lado et al., 2008). Sill and nugget values for LnC were 0.26 and 0.10, respectively (Table 4). Greater variations in the SOC stock are indicated by the sills of the variogram models, whereas, range coefficients show that the SOC stock was spatially continuous for certain distance and then became constant. Variogram models show that the spatial structure for LnC is present in a range up to 16 km. Range of the residuals$_{MLR}$ and residuals$_{GWR}$ was 15 and 10 km, respectively. Smaller range value for residuals$_{GWR}$ compared to residuals$_{MLR}$ indicates that much of the structured variation in SOC stock is explained by its local relationship with the environmental variables. A study conducted in the U.K. by Lloyd (2010) also reported a shorter range for GWR residuals for estimating the monthly precipitation amount. Correlation between the residuals and the SOC stock due to the remaining

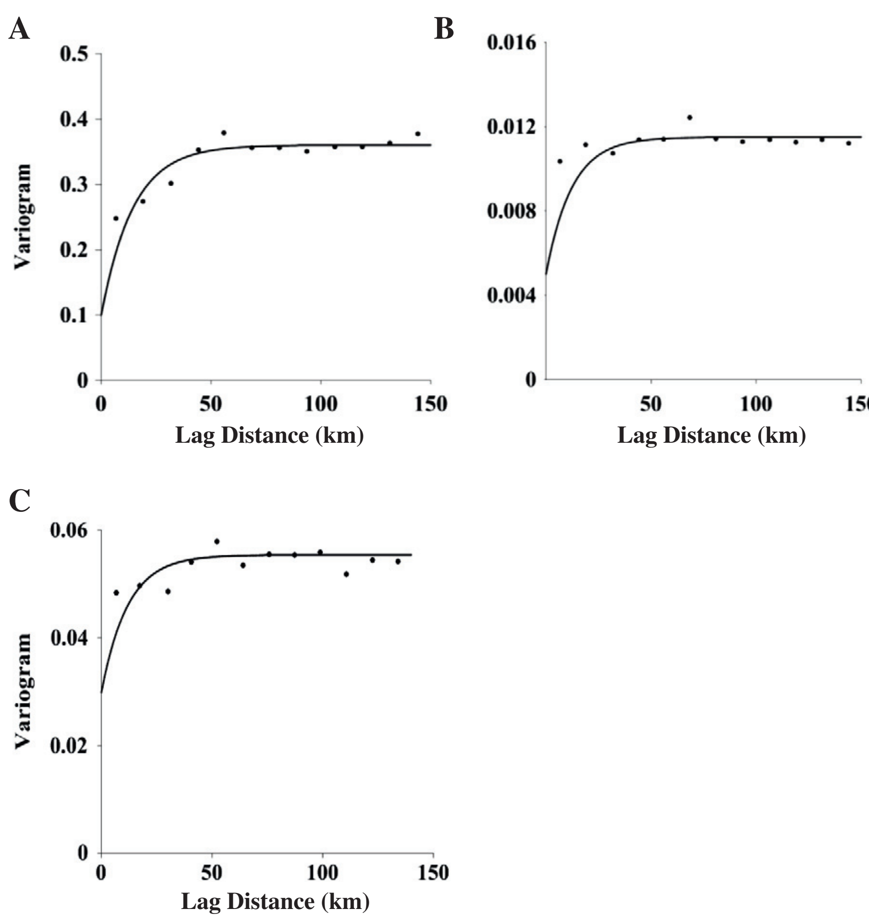

Fig. 2. Experimental variogram and fitted models of natural log-transformed soil organic carbon (A), residuals from multiple linear regression (B), and residuals from geographically weighted regression (C).

Table 4
Semivariogram models for log-transformed soil organic carbon (LnC), multiple linear regression (MLR) residuals, and geographically weighted regression (GWR) residuals.

Parameter	LnC	MLR residuals	GWR residuals
Model	Exponential	Exponential	Exponential
Range (km)	16	17	10
C_0[a] (km m^{-2})^2	0.10	0.0020	0.015
C (km m^{-2})^2	0.26	0.0095	0.040
C_0/C (%)	38	21	37

[a] C_0, nugget; C, partial sill.

Table 5
Validation indices for estimating soil organic carbon (kg m^{-2}) stock using regression kriging (RK), and geographically weighted regression kriging (GWRK) approaches.

Method	R^2	MEE[a]	RMSE	MAEE
		kg m^{-2}		
RK	0.23	−0.21	4.61	3.20
GWRK	0.36	0.11	2.61	2.34

[a] MEE, mean estimation error; MAEE, mean absolute estimation error; RMSE, root mean square error.

unexplained variability in the dependent variable (SOC stock in this study) was minimized by kriging the residuals using the GWRK approach. The diameter of the interpolation window in the interpolation for the present study was kept within the range values of the variogram models.

The GWRK approach yielded minimum prediction errors (MEE = 0.11 kg m^{-2}; RMSE = 2.61 kg m^{-2}; MAEE = 2.34 kg m^{-2}; Table 5; Figure 3) and provided good and reasonable results compared to those estimated from the RK. The RMSE for the GWRK (2.61 kg m^{-2}) was 43% lower when compared to the RK (4.61 kg m^{-2}). A similar trend was observed for MEE and MAEE values. The difference between RMSE and MAEE was the lowest (0.27 kg m^{-2}) for the GWRK, indicating small prediction errors. Residuals obtained from GWR are not randomly distributed across the space, but show some trend which can be captured by the GWRK approach. Data show that RK and GWRK reduced the prediction errors after adding the stochastic part (interpolated residuals) to MLR and GWR, respectively. The MLR and GWR produced RMSE values of 4.91 and 3.72 kg m^{-2}, respectively, which is 7 and 43% higher compared to those obtained from the RK (4.61 kg m^{-2}) and GWRK (2.61 kg m^{-2}) approaches (data not shown). Global regression (i.e. MLR) produced the highest RMSE values compared to other approaches, indicating that this approach may under or over estimate the SOC stock of the region. Residuals of MLR were randomly distributed across the space. A regional study conducted by Mishra et al. (2010) showed that the GWR is the best estimation approach when compared to RK and MLR. However, GWRK was the best approach compared to GWR alone for the present study. Therefore, total SOC stock for Pennsylvania was estimated using the GWRK.

Fig. 4 shows the maps of SOC stock created using RK and GWRK approaches generated from 702 data points. Both these approaches incorporated the kriged residuals. The SOC stock for 0–1.0 m depth, estimated with GWRK approach, ranged from 3.20 to 20.9 kg m^{-2} with a mean value of 8.91 kg m^{-2} (Table 6; Figure 4). The total estimated SOC stock ranged between 1.12 and 1.18 Pg using GWRK. Compared to GWRK, the RK approach exhibited a 5.8% lower average SOC stock (Table 6). The trend in SOC stock estimated with both the approaches was consistent.

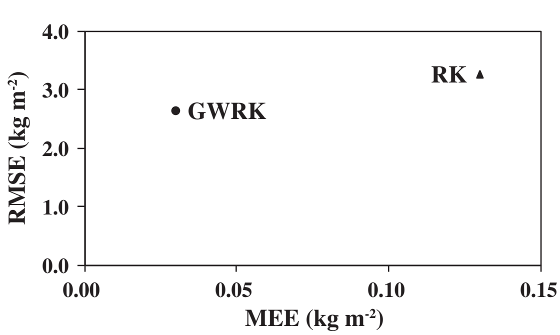

Fig. 3. Plot of estimation root mean square error (RMSE) vs. mean estimation error (MEE) for the RK and GWRK.

3.3. Physiographic distribution of SOC stock

The higher SOC stocks are stored in the northwestern part of the state and lower in the southeastern part (Figure 4). Elevated values of the stock in the northwestern region are probably due to higher rainfall and lower mean temperature. The MAP and MAAT of these areas range from 800 to 1725 mm, and 6–14 °C, respectively, with elevation ranging from 300 to 800 m. SOC stocks increased with increasing elevation above sea level probably because of higher rainfall and lower mean annual temperatures (Davidson and Janssens, 2006; Phachomphon et al., 2010).

Fig. 5A shows SOC stock stored in different MLRAs of Pennsylvania. Stock ranged from 4.60 to 15.9 kg m^{-2} with the highest value estimated in MLRA 127 (Eastern Allegheny Plateau and Mountains; 31%) and the least in MLRA 101 (Ontario-Erie Plain and Finger Lakes Region; 0.1%). MLRAs 127 and 147 (Northern Appalachian Ridges and Valleys) with an area of 56,808 km^2 (i.e. 48% of the total study area) store 51% of the total SOC stock (1.12 Pg; Figure 5A and B) of Pennsylvania. Other MLRAs having higher SOC stock are 140 (Glaciated Allegheny Plateau and Catskill Mountains; 0.18 Pg), 126 (Central Allegheny Plateau; 0.08 Pg), and 148 (Northern Piedmont; 0.06 Pg; Figure 5A).

About 57% of the total area of MLRA 127 and 54% of MLRA 147 is present in Pennsylvania (Figure 5B). The dominant soil orders of these MLRAs are Inceptisols, Ultisols, and Alfisols (only present in MLRA 147) with some suborders such as udalf (Alfisols) and udept (Inceptisols) that have a udic moisture regime. A udic moisture regime is common to soils of humid climates that have well-distributed rainfall. Soils of these MLRAs have a mesic (8–15 °C) or frigid (<6 °C) soil temperature regime and are generally moderately deep to very deep. Higher moisture and lower temperature in these MLRAs also contributed to higher SOC stock. Furthermore, the most frequent land use types present in MLRAs 127 vs. 147 are forests (68 vs. 48%), cropland (6 vs. 19%) and shrubs (6 vs. 12%). Predominantly, soils present in MLRAs 127 and 147 are undisturbed (i.e. forest soils) which may have increased the stock of these areas.

A comparison of different land uses of Pennsylvania revealed that forests stored the highest SOC stock (64%), followed by cropland (22%), developed land (10%), wetland (2.3%), shrubs (2%), and barren land (0.4%; Table 7). Pennsylvania's forests (0.75 Pg) store about 3 times higher SOC stock compared to that of cropland (0.26 Pg), and account for 64% of the total SOC stock because of its large area (about 60% of the total state area). Additionally, tillage-induced perturbations decrease soil aggregation in cropland soils and reduce the physical protection of soil organic matter (SOM). As a result,

Table 6
Descriptive statistics of the estimated soil organic carbon (SOC; kg m^{-2}) stock.

Variables	Regression kriging	Geographically weighted regression kriging
Mean (kg m^{-2})	8.39	8.91
Minimum (kg m^{-2})	2.69	3.20
Maximum (kg m^{-2})	14.6	20.9
Standard Deviation (kg m^{-2})	1.96	2.83

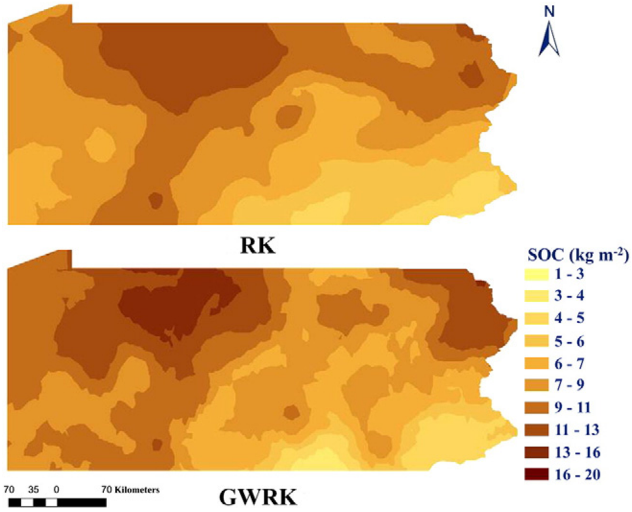

Fig. 4. Maps of estimated soil organic carbon (SOC; kg m^{-2}) stock for the soils of Pennsylvania using regression kriging (RK) and geographically weighted regression kriging (GWRK) approaches.

cropland soils generally have lower SOC stock compared to their potential capacity (Lal, 2005). In contrast, tree roots build up the SOC compared to grasslands and pastures (Kumar et al., 2010), and hence SOC stock in forest land use is generally higher compared to that in cropland or pastures. Wetlands of Pennsylvania store about 2.3% (0.027 Pg) of the total (1.12 Pg) SOC stock. Prolonged saturation of soils in wetlands have slower decomposition rates of litter and organic matter as compared to other land uses, which enhances the SOC stock in wetland soils (Qualls and Richardson, 2008). Larger spatial variations in the SOC stock are caused due to diversified climate and management practices across the state.

3.4. Uncertainties in the present study

There are a few uncertainties associated with the present study that are in need of improvement, and few of them are: (i) SOC stock is usually calculated from the bulk density values that are derived from pedotransfer functions (PTFs) (Bellamy et al., 2005; Calhoun et al., 2001). These PTFs, however, sometimes under or overestimate the bulk density when used in environments other than those in which these functions are developed (De Vos et al., 2005) because

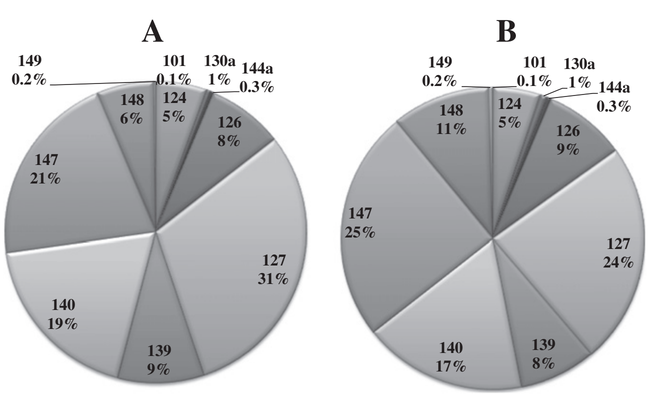

Fig. 5. Soil organic carbon stock (Pg; A), and the area (% of the total study area; B) occupied by different major land resource areas: 101, Ontario-Erie Plain and Finger Lakes Region; 124, Western Allegheny Plateau; 130a, Northern Blue Ridge; 144a, New England and Eastern New York Upland, Southern Part; 126, Central Allegheny Plateau; 127, Eastern Allegheny Plateau and Mountains; 139, Lake Erie Glaciated Plateau; 140, Glaciated Allegheny Plateau and Catskill Mountains; 147, Northern Appalachian Ridges and Valleys; 148, Northern Piedmont; 149, Northern Coastal Plain.

Table 7
Minimum (Min), maximum (Max), mean, standard deviation (SD) of soil organic carbon (SOC) stock stored in different land uses of study area.

Land Use	Area		Min	Max	Mean	SD	Total SOC	
	km^2	%	kg m^{-2}				Pg	%
Wetland	2696	2.3	4.10	15.6	8.90	2.40	0.027	2.30
Barren	489	0.4	4.00	16.4	9.30	3.00	0.005	0.40
Shrubs	1812	1.6	6.75	16.6	11.8	2.10	0.024	2.00
Cropland	28,323	24.5	3.80	16.5	8.10	2.30	0.255	21.7
Forest	69,928	60.4	4.20	16.5	10.0	2.40	0.752	63.9
Developed	12,562	10.8	3.90	16.0	7.80	2.30	0.114	9.70
Total	115,810	100					1.18	100

of differences in soil types, land use management, vegetation, and climate across the region, (ii) data were extracted from a historical database. Since SOC is sensitive to changes in vegetation cover, harvest of biomass residues in croplands and forests, and to all kinds of mechanical soil disturbances such as ploughing (Schrumpf et al., 2008). The current status of SOC stock might have changed which may provide some estimation errors finally, (iii) estimations of the SOC stock is limited to 1.0-m depth which underestimates the stock beyond this depth, since significant amount of stock usually present beyond 1.0-m depth.

4. Conclusions

In this study, some of the limitations of previous studied approaches that assumed relationship between environmental variables and the SOC stock to be uniform or stationary throughout a given region, were addressed. Therefore, the GWRK, a local approach that emphasizes the spatial non-stationarity variability coupled with spatial autocorrelation, was used to estimate the SOC stock up to 1-m for Pennsylvania.

Two geostatistical hybrid approaches (GWRK and RK) were studied and compared in this study for estimating the SOC stock. Results show that the performance of GWRK yielded less prediction errors, and outperformed the RK. The GWRK interpolated spatial distribution map of SOC stock showed clear influences of the environmental variables, and it reduced the smoothing effect problem of RK approach. Further, GWRK can provide an alternative to other global models when estimating non-stationarity relationships, especially in SOC stock which is highly variable spatially. It is unlikely that a single estimation approach and the same environmental variables are optimal for all the regions, and hence, it is obvious that some sort of uncertainties will always be present in the estimations. Similarly, despite some uncertainties found in the present study, the GWRK presented a realistic picture of the spatial distribution of the SOC stock. This approach can provide better estimations at larger scale, provided there is a strong correlation between environmental variables and the SOC stock, and residuals are spatially autocorrelated. In conclusion, the GWRK outperforms the global models by explicitly addressing spatial dependency and represents a new and more accurate way of estimating the SOC stock across the regional scale.

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